

Curriculum Vitae

Osamu SANDANBATA (Name in passport: Osamu SANDAMBATA)

Assistant Professor

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Research Interest

- Dynamics of volcanic calderas interacting intra-caldera fault systems and a magma reservoir
- Numerical modeling and analysis of tsunami and seismic waves
- Mechanical modeling of volcanic phenomena

Positions

2023.05 – Present	Assistant Professor, Earthquake Research Institute, The University of Tokyo
2023.04 – 2023.04	Postdoctoral Researcher, Earthquake Research Institute, The University of Tokyo
2020.04 – 2023.03	Research Fellow of Japan Society for the Promotion of Science (PD), Earthquake and Tsunami Research Division, National Research Institute for Earth Science and Disaster Resilience
2017.04 – 2020.03	Research Fellow of Japan Society for the Promotion of Science (DC1)

Education

2017.04 – 2020.03	Ph.D., Earthquake Research Institute, Department of Earth and Planetary Science, The University of Tokyo
2015.04 – 2017.03	M.S., Earthquake Research Institute, Department of Earth and Planetary Science, The University of Tokyo
2011.04 – 2015.03	B.S., Department of Earth and Planetary Science, The University of Tokyo

Visits

2023.12	Visiting Researcher at Seismological Laboratory, California Institute of Technology
2022.04 – 2022.07	Visiting Scholar at Geophysics Department, Stanford University
2019.07 – 2019.09	Visiting Researcher at Seismological Laboratory, California Institute of Technology

Publications

Submitted paper (not peer-reviewed)

1. N/A

Published Articles

1. **Sandanbata, O.**, & Saito, T. (2024b). Segmented Trapdoor Fault in Kita-Ioto Caldera, Japan: Insights From Millimeter Tsunami Waves Captured by an Array Network of Ocean Bottom Pressure Gauges. *Journal of Geophysical Research: Solid Earth*, 129(12), e2024JB029755. <https://doi.org/10.1029/2024JB029755>
2. Takemura, S., Kubota, T., & **Sandanbata, O.** (2024). Successive tsunamigenic events near Sofu Seamount inferred from high-frequency teleseismic P and regional T waves. *Journal of Geophysical Research: Solid Earth*, 129, e2024JB029746. <https://doi.org/10.1029/2024JB029746>
3. Kubota, T., **Sandanbata, O.**, Saito, T., & Matsuzawa, T. (2024). Accelerating seafloor uplift of submarine caldera near Sofugan volcano, Japan, resolved by distant tsunami recordings. *Geophysical Research Letters*, 51(12). <https://doi.org/10.1029/2024gl108415>
4. **Sandanbata, O.**, Satake, K., Takemura, S., Watada, S., Maeda, T., & Kubota, T. (2024). Enigmatic tsunami waves amplified by repetitive source events near Sofugan volcano, Japan. *Geophysical Research Letters*, 51, e2023GL106949. <https://doi.org/10.1029/2023GL106949>
5. **Sandanbata, O.**, & Saito, T. (2024). Quantifying magma overpressure beneath a submarine caldera: A mechanical modeling approach to tsunamigenic trapdoor faulting near Kita-Ioto Island, Japan. *Journal of Geophysical Research: Solid Earth*, 129, e2023JB027917. <https://doi.org/10.1029/2023JB027917>
6. **Sandanbata, O.**, Watada, S., Satake, K., Kanamori, H., & Rivera, L. (2023). Two volcanic tsunami events caused by trapdoor faulting at a submerged caldera near Curtis and Cheeseman Islands in the Kermadec Arc. *Geophysical Research Letters*, 50, e2022GL101086. <https://doi.org/10.1029/2022GL101086>
7. **Sandanbata, O.**, Watada, S., Satake, K., Kanamori, H., Rivera, L., & Zhan, Z. (2022). Sub-decadal volcanic tsunamis due to submarine trapdoor faulting at Sumisu caldera in the Izu–Bonin Arc. *Journal of Geophysical Research: Solid Earth*, 127, e2022JB024213. <https://doi.org/10.1029/2022JB024213>
8. Kubo, H., Kubota, T., Suzuki, W., Aoi, S., **Sandanbata, O.**, Chikasada, N., & Ueda, H. (2022). Ocean-

wave phenomenon around Japan due to the 2022 Tonga eruption observed by the wide and dense ocean-bottom pressure gauge networks. *Earth Planets Space* **74**, 104. <https://doi.org/10.1186/s40623-022-01663-w>

9. Kubota, T., Saito, T., Chikasada, N. Y., & **Sandanbata, O.** (2021). Meteotsunami observed by the deep-ocean seafloor pressure gauge network off northeastern Japan, *Geophysical Research Letters*, 48(21) e2021GL094255. <https://doi.org/10.1029/2021GL094255>
10. Lai, V. H., Zhan, Z., Brissaud, Q., **Sandanbata, O.**, & Miller, M. S. (2021). Inflation and Asymmetric Collapse at Kilauea Summit during the 2018 Eruption from Seismic and Infrasound Analyses, *Journal of Geophysical Research: Solid Earth*, 126(10), e2021JB022139. <https://doi.org/10.1029/2021JB022139>
11. **Sandanbata, O.**, Kanamori, H., Rivera, L., Zhan, Z., Watada, S., & Satake, K. (2021). Moment tensors of ring-faulting at active volcanoes: Insights into vertical-CLVD earthquakes at the Sierra Negra caldera, Galápagos Islands, *Journal of Geophysical Research: Solid Earth*, 126(6), e2021JB021693. <https://doi.org/10.1029/2021JB021693>
12. **Sandanbata, O.**, Watada, S., Ho, T-C., & Satake, K. (2021). Phase delay of short-period tsunamis in the density-stratified compressible ocean over the elastic Earth, *Geophysical Journal International*, 226(3), 1975–1985. <https://doi.org/10.1093/gji/ggab192>
13. Saito, T., Kubota, T., Chikasada, N. Y., Tanaka, Y., & **Sandanbata, O.** (2021). Meteorological tsunami generation due to sea-surface pressure change: Three-dimensional theory and synthetics of ocean-bottom pressure change, *Journal of Geophysical Research: Oceans*, 126(5), e2020JC017011. <https://doi.org/10.1029/2020JC017011>
14. Heidarzadeh, M., Ishibe, T., **Sandanbata O.**, Muhari, A., & Wijanarto, A. B. (2020). Numerical modeling of the subaerial landslide source of the 22 December 2018 Anak Krakatoa volcanic tsunami, Indonesia, *Ocean Engineering*, 195, 106733. <https://doi.org/10.1016/j.oceaneng.2019.106733>
15. Wang, Y., Satake, K., **Sandanbata, O.**, Maeda, T., & Su, H. (2019). Tsunami data assimilation of cabled ocean bottom pressure records for the 2015 torishima volcanic tsunami earthquake, *Journal of Geophysical Research: Solid Earth*, 124(10), 10413–10422. <https://doi.org/10.1029/2019JB018056>
16. **Sandanbata, O.**, Shiobara, H., Kusumoto, S., Kim, H. J., Oba, A., Liu, Q., Ueda, T., Ogawa, M., Takano, K., Kotobuki, I., & Wang, Y. (2018). Equipment of Miniature Instruments to Measure Tsunami Waves in an Experimental Tank (in Japanese), *Technical Research Report, Earthquake Research Institute, the University of Tokyo*, 24, 29–34. http://www.eri.u-tokyo.ac.jp/GIHOU/archive/24_029-034.pdf
17. **Sandanbata, O.**, Watada, S., Satake, K., Fukao, Y., Sugioka, H., Ito, A., & Shiobara, H. (2018). Ray Tracing for Dispersive Tsunamis and Source Amplitude Estimation Based on Green's Law: Application to the 2015 Volcanic Tsunami Earthquake Near Torishima, South of Japan, *Pure and Applied Geophysics*, 175, 1371–1385. <https://doi.org/10.1007/s00024-017-1746-0>
18. Fukao, Y., **Sandanbata, O.**, Sugioka, H., Ito, A., Shiobara, H., Watada, S., & Satake, K. (2018). "Mechanism of the 2015 volcanic tsunami earthquake near Torishima, Japan." *Science advances* 4, no. 4, eaao0219. <https://doi.org/10.1126/sciadv.aao0219>

19. **Sandanbata, O.**, Obara, K., Maeda, T., Takagi, R., & Satake, K. (2015). Sudden changes in the amplitude-frequency distribution of long-period tremors at Aso volcano, southwest Japan, *Geophysical Research Letters*, 42, 10,256–10,262. <https://doi.org/10.1002/2015GL066443>

Honors and Awards

1. AGU's Editor Highlight for Takemura *et al.* (2024), 2025.
2. The University of Tokyo Excellent Young Researcher, The University of Tokyo, 2023.
3. Early-Career Researcher Plenary Speaker, IAVCEI 2023 Scientific Assembly, 2023.
4. Student Presentation Award, Seismological Society of Japan Fall Meeting 2018, 2018.
5. Outstanding Student Presentation Award, JpGU Annual Meeting 2018, 2018.
6. Outstanding Student Presentation Award, JpGU-AGU Joint Meeting 2017, 2017.
7. Best Student Poster Award, Asia Oceania Geoscience Society 13th Annual Meeting, 2016.
8. Outstanding Student Presentation Award, JpGU Annual Meeting 2016, 2016.

Invited Talk

1. **Sandanbata, O.**, Unveiling hidden volcanic activity of submarine calderas based on tsunami and seismic records, Seminar of Center for Deep-Surface Coupling of Earth, *Seoul National University*, South Korea. September 2024.
2. **Sandanbata, O.**, Exploring unknown submarine volcanic activity using tsunami and seismic wave records (in Japanese), Seismology Special Seminar, *Hirosaki University*, Japan. November 2023.
3. **Sandanbata, O.**, Unusual volcanic tsunamis caused by trapdoor faulting at submarine calderas, Brown Bag Seminar of Seismological Laboratory, *California Institute of Technology*, US. April 2022.
4. **Sandanbata, O.**, Unusual volcanic tsunamis caused by trapdoor faulting at submarine calderas, Bullard Laboratories Seminar, *the University of Cambridge*, UK. March 2022.
5. **Sandanbata, O.**, Investigations of ring-faulting at active calderas: Case studies at Sierra Negra caldera in Galápagos, and Smith caldera in Japan, Volcano Group Meeting, *University of Leeds*, UK. April 2021.
6. **Sandanbata, O.**, Abnormal tsunamis caused by trapdoor faulting repeating at submarine volcanic calderas, Geophysics Seminar, *Stanford University*, US. April 2021.

Fellowships and Grants

Fellowship

1. Japan Society for the Promotion of Science (JSPS) Research fellow (PD), 2020.04 – Present.
2. Japan Society for the Promotion of Science (JSPS) Research fellow (DC1), 2017.04 – 2020.03.

Grants

1. Grant-in-Aid for Early-Career Scientists, #24K17141 (2024.04.01–2028.03.31: JPY4,810,000)

2. The Sasakawa Scientific Research Grant, The Japan Science Society, #2023–2031 (2023.04.01–2024.02.10: JPY1,500,000)
3. Grant-in-Aid for JSPS Fellows, #20J01689 (2020.04.01–2023.03.31: JPY4,810,000)
4. Grant-in-Aid for JSPS Fellows, #17J02919 (2017.04.01–2020.03.31: JPY4,160,000)

Thesis Titles

Ph.D.	Physical mechanism of volcanic tsunami earthquakes repeating at submarine volcanoes. (Supervisor: Prof. Kenji Satake)
M.S.	Ray Tracing for Dispersive Tsunamis and Estimation of Initial Sea-surface Deformation from Array Data: Application to the 2015 Volcanic Tsunami Earthquake near Smith Caldera, South of Japan. (Supervisor: Prof. Kenji Satake)
B.S.	Step-wise changes in the amplitude-frequency distribution of long-period tremors at Aso volcano (in Japanese). (Supervisor: Prof. Kazushige Obara)

Teaching Experiences

2023.09	Summer School of Earthquake Research Institute, The University of Tokyo.
2023.04 – 2023.06	Senior Research Project in Earth and Planetary Physics, The University of Tokyo.
2017.06 – 2017.07	Tutor for two international internship students of <i>University of Tokyo Research Internship Program</i> , The University of Tokyo.
2017.04 – 2017.09	Teaching Assistant, <i>Seismic Wave Theory</i> , The University of Tokyo.
2016.09 – 2017.03	Mentor for an international student in master program, The University of Tokyo.
2016.06 – 2016.07	Tutor for two international internship students of <i>University of Tokyo Research Internship Program</i> , The University of Tokyo.
2015.06 – 2015.07	Tutor for two international internship students of <i>Sakura Science Plan</i> , Earthquake Research Institute, The University of Tokyo.
2014.11	Tutor for three international internship students of <i>Sakura Science Plan</i> , Earthquake Research Institute, The University of Tokyo.

Outreach Experiences

2015.08, 2016.08, and 2018.08	Open-laboratory experiment of tsunami generation/propagations in a miniature tank, Earthquake Research Institute, The University of Tokyo.
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